Automated End-to-End 3-Axis CNC Machining: Bi-Level Task-and-Motion Planning, Differentiable Cutting Simulation, and Trajectory-Policy Co-Optimization

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Abstract—Advances in fused-deposition 3D printing have made additive fabrication accessible to non-experts: a user drops a digital model into slicer software, which automatically plans and executes trajectories in $\mathbb{R}^4$ $(x, y, z, \text{extrude})$. That accessibility has not transferred to 3-axis Computer Numerical Control (CNC) milling. Although 3-axis CNC trajectories inhabit a superficially similar continuous action space $\mathbb{R}^4$ $(x, y, z, \text{spindle/cutting speed})$, subtractive machining requires first solving a combinatorial upstream task-planning problem—selecting raw stock alloy and dimensions, $SO(3)$ setup orientation and multi-setup workholding, multi-class cutting/drilling/chamfering/threading tool cribs, feature-bound toolpath topologies, and narrow physically admissible bands of feeds and speeds—followed by long-horizon trajectory optimization through discontinuous Boolean material removal. Today, this hierarchy demands hours of manual configuration by expert CNC programmers. We challenge this paradigm with DiffCAM, an MIT-licensed, end-to-end automated 3-axis CNC machining framework that unifies (1) bi-level task-and-motion planning (TAMP) over arbitrary ISO 10303-21 .step B-Rep and .stl CAD solids, 6-sided $SO(3)$ setup scoring with automated Op1 (G54) / Op2 (G55) soft-jaw flipping, a 673-record US/Metric clean-room tool catalog with customizable shop tool cribs, and a 6-constraint feeds_and_speeds physics solver; (2) classical clean-room CAM trajectory initialization across datum facing, Euclidean-EDT 2.5D roughing/rest-finishing, trochoidal clearing, inverse-offset 3D ball-endmill scallop surfacing, helical boring, G83 peck drilling, chamfering, and climb thread milling; and (3) dual-mode differentiable cutting simulation combining 3D swept-cylinder signed distance fields (SDFs) under Straight-Through Estimator (STE) continuation with chronological 2.5D cutter-workpiece engagement (CWE) chip-thinning mechanics and exact G187 junction-blending gradients. Across 16 single-tool benchmark tasks ($B = 64$ simulator calls), smooth-to-hard STE continuation cuts cycle time by $\mathbf{42.0 \pm 13.5}$% at 100% Boolean feasibility (4× evolutionary search), while first-order DiffPolicy-SHAC doubles the sample efficiency of zero-order Hard-PPO (**37.0**% vs. **18.5**%). Under matched-speed and optimal-chipload CWE evaluation (Exp5–Exp5b), DiffCAM-Chipload maintains **97.7**% of engaged cutting inside the optimal undeformed chip-thickness band (h_{ex} RMSE 0.0053 mm), and inference-time trajectory overfitting from zero-shot SHAC (3.5 ms) or CAM seeds converges in 0.48–1.04 s. Finally, across 4 multi-alloy .stl parts (Exp6) and all 10 arbitrary ISO 10303-21 .step parts of the TITAN-1M–10M aerospace benchmark (Exp7, spanning 6–11 tools/part, G54+G55 flips, 3D B-spline domes/airfoils, and internal threads), DiffCAM achieves **100**% safety pass (**0.000** mm^3 gouge), **36.3**% fair matched-speed savings, and **41.6**% optimal-chipload cycle-time reduction across 60/60 compiled 6-dialect NC programs.

I. Introduction

Roboticists in the past have often dismissed the robotic challenges of manufacturing because the factory domain appears relatively well-defined and structured rather than addressing open-ended, general-purpose manipulation tasks. Yet creating physical objects from digital designs via machine tools has historically required deep artistic and technical expertise. Advances in desktop 3D printing have demonstrated that technical expertise in manufacturing processes is not inherently required to manifest digital objects in the physical world: a user no longer needs to understand toolpaths, kinematics, or robotics, and can instead simply drop a digital object into a 3D printer's slicer software, which automatically finds and executes the toolpaths required to extrude plastic in the shape of the part.

However, the plug-and-play accessibility of 3D printing software has not transferred to more complex manufacturing processes, such as 3-axis CNC milling machines. At the continuous motion level, the instantaneous action space for a typical fused-deposition modeling (FDM) 3D printer is $\mathbb{R}^4$ $(x, y, z, \text{extrude})$, and the action space for 3-axis CNC machining trajectories is superficially similar—$\mathbb{R}^4$ $(x, y, z, \text{spindle/cutting speed})$. Relative to 3D printing, CNC machining allows the fast and repeatable creation of strong, precision parts out of both common and exotic structural alloys. However, because machining is a subtractive process—in which a rotating cutter permanently removes solid metal and every pass irreversibly alters downstream cutter engagement [1, 22]—much more upstream planning is required prior to choosing a continuous trajectory:

1) Stock, Orientation, and Multi-Setup Workholding: One must first select a piece of raw material, called stock (including material alloy and billet dimensions), choose the 3D orientation of the stock inside the CNC work envelope, and determine how it is rigidly clamped down against cutting forces, called workholding (such as parallel vise jaws or custom

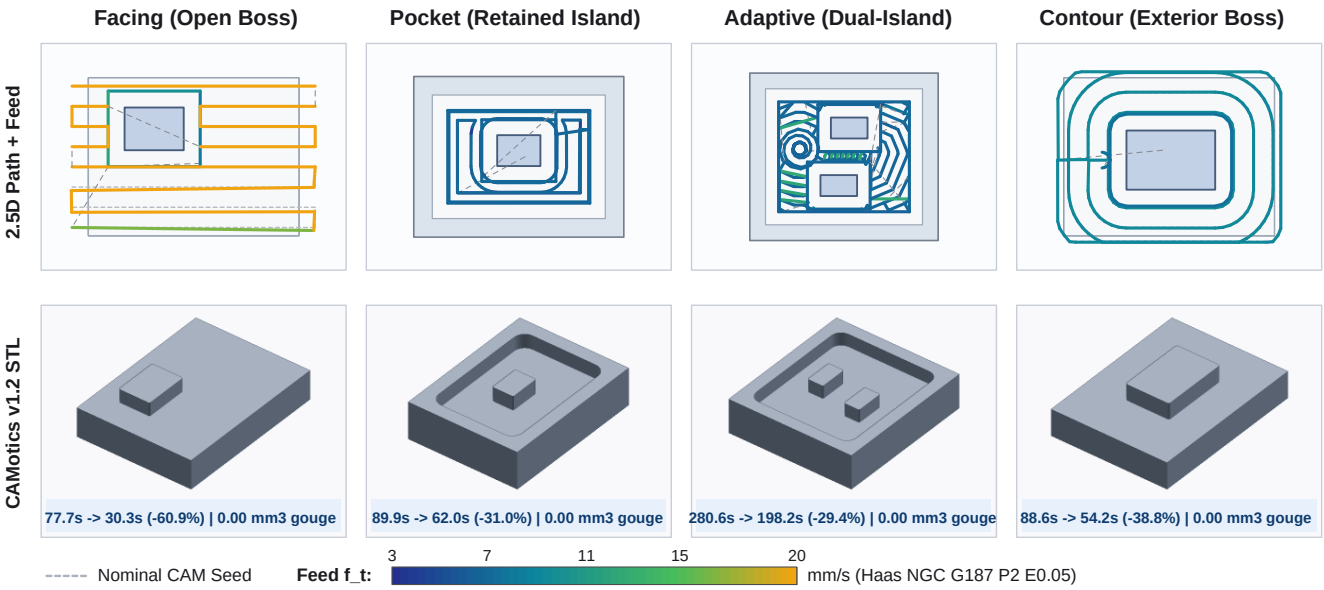


Fig. 1. DiffCAM joint 2.5D toolpath and feedrate co-optimization certified in CAMotics v1.2.0. Top row: Representative held-out test parts across four modular MIT clean-room CAM strategy families showing stock billets, retained islands/bosses, dashed gray nominal CAM seeds (37–370 segments), and optimized STE-Continuation-Adam trajectories colored by per-segment feedrate $f_t \in [2.8, 20.0]$ mm/s under Haas Next-Gen Control corner blending (G54, G187 P2 E0.05). Bottom row: 4×-MSAA isometric renderings of the closed, watertight binary STL meshes reconstructed directly by containerized CAMotics v1.2.0 (0.25 mm grid) from posted Haas NGC G-code, achieving 29.4%–60.9% cycle-time reductions on these four parts (39.5% mean across all 8 test parts) with 0.00 mm^3 target gouge, 0.00 mm^3 residual, and exact STL volume identity with nominal CAM.

soft jaws). Whenever underside pockets, counterbores, or blind holes are occluded from above, the part must also be flipped across multiple work offsets (G54, G55) without colliding with the vise jaws.

2) Multi-Class Tool Cribs and Operation Topologies: Whereas 3D printers typically operate with a single nozzle size and one filament corresponding to a single operation (extruding plastic), cost- and time-efficiently removing metal from stock requires a library of tools spanning distinct classes of operations: cutting tools (face, flat, bull-nose, ball, and chamfer end mills), hole-making tools (spot drills, twist drills, countersinks, reamers), threading tools (taps, thread mills), and spindle touch probes. Each tool corresponds to specialized parameterized toolpaths: drills plunge in a single motion or execute incremental pecks and retract to evacuate chips before plunging deeper (G83); cutting tools slice a thin datum layer off the top of the stock via a facing path, hog out bulk pockets rapidly via roughing or trochoidal clearing [9, 16], rest-mill tight corners left behind by larger cutters, or finish sculptured 3D domes and airfoils via ball-endmill scallop passes [3]. Moreover, any real shop only stocks a finite subset of tools (its tool crib), requiring automatic fallback strategies such as helical bore interpolation when an exact drill size is unavailable.

3) Narrow Admissible Feeds and Speeds: Tools and toolpaths admit only a narrow physical band of configurations—specifically the rotational speed of the tool relative to linear motion against the workpiece, known as feeds and speeds [1, 14]. Feeding below minimum chip thickness causes rubbing and work-hardening; feeding too aggressively exceeds spindle power, core shear torque, or cantilever deflection limits and fractures the cutter.

The combination of stock material, workholding, tool selection, toolpath choice, and feeds and speeds creates a massive hierarchical planning space that must be solved before finding an optimal trajectory under the conditions of the plan. Today, people solve the complexity of CNC machining in a predominantly manual way: expert CNC programmers spend many hours in visual CAM tools manually configuring stock, workholding, US or metric tool-library assemblies, feature-associated parameterized toolpaths, and handbook- or calculator-derived feeds and speeds.

We challenge this paradigm and propose that 3-axis CNC machining should be end-to-end solvable in an automated way—at a quality level that currently requires hours of expert programmer time—by coupling task-level setup and multi-tool planning prior to trajectory optimization with classical CAM trajectory initialization, physics-grounded feeds and speeds, and differentiable cutting simulation. Achieving this requires resolving three core algorithmic challenges:

1) Bi-Level Coupling Between Discrete Tool Cribs and Persistent Stock Geometry: Upstream tool choices

govern both automatic-tool-changer (ATC) swap overhead and the exact evolving in-process workpiece (IPW) encountered by every downstream rest-milling, 3D surfacing, and hole-making tool across single or dual setups.

2) Discontinuous Boolean Subtraction, Radial Chip Thinning, and Corner Blending: Exact Boolean subtraction $\phi_{t+1} = \max(\phi_t, -\psi_{\text{tool}})$ [20, 23] has zero gradients in air and kinks on cut boundaries, naive smooth surrogates suffer volumetric drift [15, 21], relative waypoint deltas accumulate $O(T)$ drift, and naive feed boosting along curved polylines triggers severe CNC junction deceleration penalties (G64/G187) [5, 13] or distorts actual undeformed chip thickness h_{ex} under partial radial immersion [14].
3) Amortized RL Generalization vs. Per-Part Test-Time Optimality: Offline-trained RL policies infer in milliseconds on unseen parts but suffer finite-capacity approximation error, motivating fast test-time trajectory overfitting through analytical simulator gradients.

To address this full stack, we present DiffCAM (Figs. 1 and 2). Our contributions are:

- Arbitrary STEP/STL Multi-Setup Bi-Level TAMP & Configurable Tool Cribs: We build an MIT clean-room catalog of 673 US/Metric tools and holders (StandardToolLibrary), a 6-constraint feeds_and_speeds physics engine, an ISO 10303-21 .step B-Rep + .stl feature recognizer (cam.step_brep), and a bi-level TAMP planner (cam.tamp_planner) supporting 6-sided $SO(3)$ setup orientation, Op1 (G54) + Op2 (G55) soft-jaw flipping, crib-constrained fallback synthesis, and persistent-IPW propagation across 10 operation classes.
- Dual-Mode Differentiable Cutting Simulation with Exact Junction-Blending Gradients: We formulate (a) a GPU Taichi [10] 3D swept-SDF simulator with Straight-Through Estimator (STE) [2] stiffness continuation ($k : 25 \to 90$), and (b) a chronological 2.5D dexel CWE optimizer modeling Martellotti radial chip thinning, swept circular-endcap immersion, full-cutter-radius-verified G0 air shortcutting, and closed-form G187 junction-velocity gradients $\partial T_{\text{cycle}}/\partial v_k$.
- Controlled RL Benchmark and Sub-Second Inference-Time Trajectory Overfitting: Across 16 tasks and 8 methods under strict simulator-call parity, we compare first-order (STE-Continuation-Adam, DiffPolicy-SHAC [24]) vs. zero-order (Hard-PPO [19], CEM [17], CMA-ES [8]) methods, isolate fair matched-speed vs. optimal-chipload gains (Exp5), and demonstrate sub-second inference-time trajectory overfitting (Exp5b).
- End-to-End Verification on Arbitrary Aerospace STEP Solids and Live 3D Tool-Library WebGL: Beyond 48/48 single-tool 6-dialect and CAMotics v1.2.0 [4] watertight STL audits (Exp4) and 4 multi-alloy .stl parts (Exp6), DiffCAM machines all 10/10 ISO 10303-21 .step solids of the TITAN-1M–10M benchmark (Exp7) with 0.000 mm^3 gouge and 41.6% mean cycle-time savings, paired with an interactive CAD-upload, 3D tool-assembly, and 6-dialect G-code WebGL simulator.

II. Related Work

Task and Motion Planning (TAMP) and CAD Feature Recognition. Integrated TAMP couples discrete symbolic sequencing—such as grasp, mode, or tool selection—with continuous geometric feasibility and motion optimization [7, 12]. While widely studied in robotic manipulation and additive extrusion, subtractive CNC process planning has traditionally treated STEP B-Rep feature recognition [11, 22] separately from downstream toolpath generation. DiffCAM formulates 3-axis machining as a bi-level persistent-IPW TAMP problem: upstream $SO(3)$ setup orientation, multi-setup G54/G55 partitioning, and crib-constrained DAG tool selection minimize ATC-penalized machining time while propagating exact carved dexel stock across tools so downstream rest-finishing, 3D surfacing, and hole-making trajectories adapt to actual remaining metal.

Geometric CAM, Sculptured Surfacing, and Engagement Physics. Classical 2.5D pocketing uses 2D medial-axis/distance offsets [9], which suffer sharp radial immersion spikes in concave corners, motivating trochoidal clearing [16] and Z-buffer dexel [23], R-function [20], or octree [4] verification. Freeform 3D surfaces require inverse-offset cutter-location (z_{CL}) envelopes to prevent ball-endmill gouging [3]. Physically, when radial immersion $a_e < D/2$, Martellotti's kinematics [1, 14] dictates that maximum undeformed chip thickness h_{ex} thins below feed per tooth f_z, requiring feed compensation bounded by spindle power and cantilever beam deflection. Whereas prior feedrate schedulers modulate speed along a frozen toolpath [5], DiffCAM jointly optimizes continuous XY waypoints, stock-verified air shortcuts, and per-station feeds through differentiable CWE and corner-blending kinematics [13].

Differentiable Simulation, RL, and Test-Time Optimization. Differentiable simulators [6, 10] enable low-variance first-order trajectory optimization and short-horizon analytic actor-critic (SHAC) learning [24], though discontinuous contact or Boolean clipping can introduce gradient bias or variance [15, 21] relative to zero-order PPO [18, 19], CEM [17], and CMA-ES [8]. Crucially, we show that analytical simulator gradients unlock sub-second inference-time trajectory overfitting from either an amortized RL warm-start or a clean-room CAM seed—an operating regime where test-time model-free RL fails due to $\mathcal{O}(D/\sigma^2)$ gradient variance.

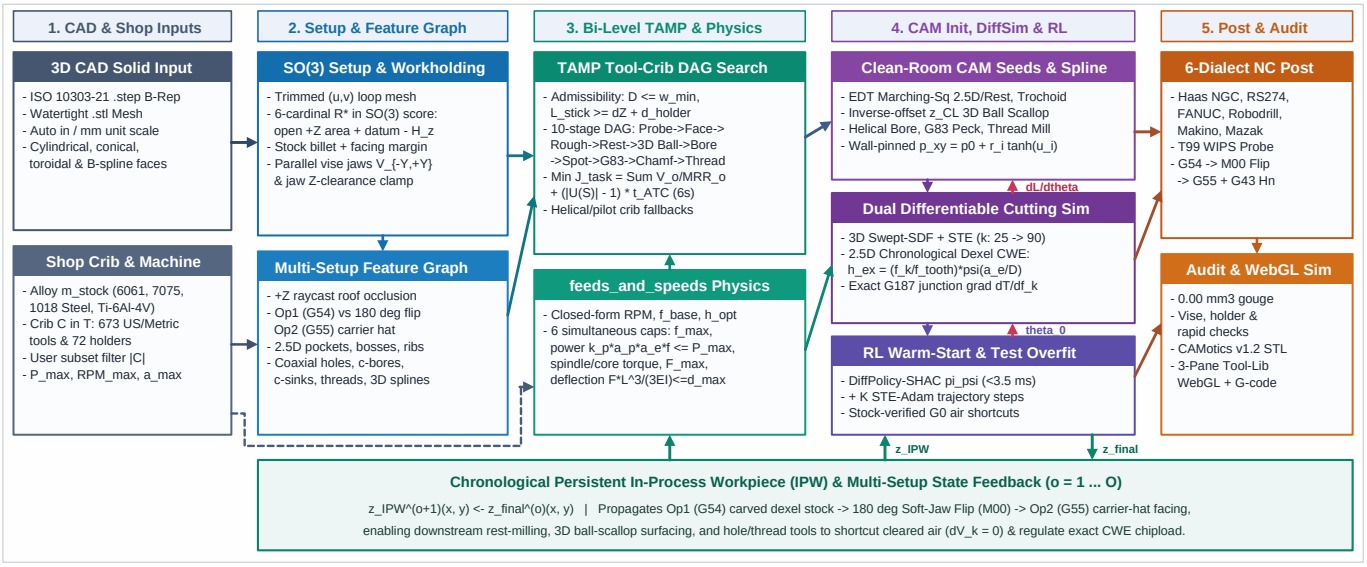


Fig. 2. DiffCAM end-to-end system architecture. Raw ISO 10303-21 .step B-Rep or .stl solids, workpiece alloy properties, and an active US/Metric Shop Tool Crib $\mathcal{C} \subseteq \mathcal{T}$ (673 clean-room tools and holders) flow left-to-right through: (2) trimmed (u, v) tessellation, 6-cardinal $SO(3)$ setup scoring, vise-jaw clamping synthesis, and raycast-occlusion feature partitioning across Op1 (G54) and flipped Op2 (G55); (3) bi-level TAMP tool-crib DAG search under ATC swap cost t_{ATC} coupled to our 6-cap feeds_and_speeds physics engine; (4) clean-room CAM trajectory initialization and wall-clearance drift-free splines co-optimized via dual-mode differentiable simulation (3D Taichi swept-SDF STE and chronological 2.5D dexel CWE with exact G187 junction gradients) and optional DiffPolicy-SHAC warm-starting; and (5) 6-controller G-code post-processing (T99 WIPS probing, G54→M00→G55), CAMotics STL verification, and 3-pane WebGL simulation. Bottom ribbon: chronological persistent-IPW feedback $z_{\text{IPW}}^{(o+1)} \leftarrow z_{\text{final}}^{(o)}$ links discrete multi-tool TAMP decisions to downstream rest-machining and air shortcutting.

III. System Architecture and Bi-Level Task-and-Motion Planning

Figure 2 illustrates the five stages of DiffCAM. To ensure complete conceptual reproducibility, Sections III-A–III-D specify every representation and algorithmic rule in the upstream TAMP and CAM-initialization pipeline, followed in Section IV by our dual-mode differentiable simulators, RL policies, and post-processors.

A. CAD Ingestion, Trimmed B-Rep Tessellation, and 6-Sided $SO(3)$ Setup Synthesis

ISO 10303-21 STEP B-Rep and STL Parsing (cam.step_brep): DiffCAM accepts either watertight binary/ASCII .stl triangle meshes or native ISO 10303-21 (AP203/AP214) .step boundary-representation solids [11]. Our clean-room STEP parser tokenizes the DATA entity graph, resolves LENGTH_MEASURE_WITH_UNIT (INCH $\times 25.4 \rightarrow$ mm vs. MILLI/METRE), and extracts every ADVANCED_FACE with its analytical or freeform carrier surface (PLANE, CYLINDRICAL_SURFACE, CONICAL_SURFACE, TOROIDAL_SURFACE, SPHERICAL_SURFACE, and rational/non-rational B_SPLINE_SURFACE_WITH_KNOTS) together with oriented outer and inner EDGE_LOOP trim curves (LINE, CIRCLE, ELLIPSE, B_SPLINE_CURVE). Each face is projected into its local 2D (u, v) parameter domain (handling 2π periodic seam unwrapping on cylinders, cones, and tori), inner hole loops are connected to the outer boundary via minimum-distance bridge cuts, and the resulting simple polygon is ear-clipped and lifted onto an interior (u, v) Steiner grid so curved surfaces are tessellated to chordal tolerance ≤ 0.08 mm while retaining exact analytical face-type, axis, radius, and semi-angle metadata on every triangle.

6-Cardinal $SO(3)$ Setup Orientation Selection: Given the tessellated solid $\mathcal{M}_0$, the planner evaluates the six cardinal axis-aligned rotations $\mathcal{R}_6 \subset SO(3)$ mapping $\{\pm X, \pm Y, \pm Z\} \mapsto +Z$ (with longest horizontal dimension aligned along machine $+X$, i.e., parallel to the vise jaws). For each candidate $\mathbf{R} \in \mathcal{R}_6$, we rasterize upper and lower heightfields at pitch $\Delta x = 0.25$ mm and score

$$\begin{aligned} S_{\text{setup}}(\mathbf{R}) = {} & w_{\text{vis}} A_{+Z}(\mathbf{R}) + w_{\text{feat}} N_{+Z}^{\text{feat}}(\mathbf{R}) \\ & + w_{\text{clamp}} A_{\text{clamp}}(\mathbf{R}) - w_{\text{ovh}} A_{\text{undercut}}(\mathbf{R}) \\ & - w_H L_z(\mathbf{R}), \end{aligned} \tag{1}$$

where A_{+Z} is upward-visible machined pocket/step/sculptured surface area, N_{+Z}^{feat} counts upward-accessible hole/bore axes, A_{clamp} measures parallel vertical side-wall datum contact along $\pm Y$ for rigid vise gripping, A_{undercut} penalizes occluded downward-facing cavities requiring a flip, and $L_z(\mathbf{R})$ penalizes tall Z-height that demands long, deflection-prone tool stickout. Selecting $\mathbf{R}^\star = \arg\max_{\mathbf{R} \in \mathcal{R}_6} S_{\text{setup}}(\mathbf{R})$ automatically reorients parts authored on side planes (e.g., mapping $+Y \mapsto +Z$ on Titan-M2, M3, M5, M6, M9, M10) without user intervention.

Stock Billet, Parallel Vise Jaws, and Multi-Setup (G54/G55) Feature Partition: In frame $\mathbf{R}^\star$, the part's

top datum is placed at $z = -\delta_{\text{top}}$ ($\delta_{\text{top}} \in [1.0, 1.5]$ mm) inside a rectangular stock billet $[x_{\min}, x_{\max}] \times [y_{\min}, y_{\max}] \times [-L_z, 0]$, where $L_z = H_{\text{part}} + \delta_{\text{top}} + \delta_{\text{grip}}$ includes a sacrificial bottom carrier-hat clamping grip $\delta_{\text{grip}} \in [2.0, 3.5]$ mm held between forbidden parallel vise jaws $\mathcal{V}_{-Y}, \mathcal{V}_{+Y}$ spanning $z \in [-L_z, z_{\text{jaw}}]$. All Op1 (G54) cutting depths are clamped to $z \geq z_{\text{jaw}} + 0.35$ mm to guarantee jaw clearance.

Coaxial cylindrical and conical face groups (or circular STL wall loops) are assembled into compound hole stacks recording bore diameter D_h, $[z_{\text{top}}, z_{\text{bot}}]$, counterbore ($D_{\text{cb}}, z_{\text{cb}}$), countersink ($D_{\text{cs}}, \alpha_{\text{cs}} \in \{82°, 90°\}$), 45° edge-chamfer chains, and internal ANSI Unified / ISO metric threads detected when bore diameter matches standard tap-drill minor diameter $D_{\text{minor}} \approx D_{\text{major}} - 1.0825\,P$ (e.g., #10-24, 1/4-20, M6×1). Freeform B-spline faces with tilted normals ($0.15 < |n_z| < 0.985$) are segmented into connected 3D sculptured regions. Crucially, to distinguish true through-holes from blind underside features, the planner casts $+Z$ rays from every feature's interior: any cavity, counterbore, or blind hole occluded from $+Z$ by overlying solid part roof ($\exists (x, y, z') \in \mathcal{M}$ with $z' > z_{\text{feat}} + \varepsilon$), or any part requiring bottom carrier-hat removal to expose underside islands/profiles, automatically triggers a second setup (Op2, work offset G55) under the 180° X-axis soft-jaw flip

$$\mathcal{F}(x, y, z) = (x,\ -y,\ -L_z - z), \tag{2}$$

inserted after a posted machine stop (M00) so the operator flips the Op1-machined profile into custom soft jaws.

B. Bi-Level TAMP: Crib-Constrained DAG Search and Persistent-IPW Coupling

How are discrete task planning and continuous motion optimization coupled? DiffCAM solves a bi-level hierarchical program over our 100% clean-room StandardToolLibrary $\mathcal{T}$ (673 total records: 601 US imperial and metric face mills, flat/bull/ball end mills, chamfer/countersink cutters, spot/twist drills, reamers, taps, single-/multi-form thread mills, and T99 touch probes, plus 72 BT30/CAT40/HSK/ISO20 holders across 15 library groups and 32 entity types):

1) Shop Tool Crib Filtering and Geometric/Reach Admissibility: Given a user-selected shop crib $\mathcal{C} \subseteq \mathcal{T}$ (e.g., US Imperial Crib, Metric Shop Crib, or an explicit subset of in-magazine tools), a tool $\tau \in \mathcal{C}$ is admissible for feature f iff: (a) its cutting diameter satisfies local feature clearance ($D \leq \min(2r_{\min}(f), w_{\min}(f) - \delta_c)$ for rest-finishing; $0.50 D_{\text{bore}} \leq D \leq 0.88 D_{\text{bore}}$ for helical bores; $D \leq 0.82 D_{\text{minor}}$ with matching pitch P for thread mills), and (b) assembly stickout clears feature depth below stock top without holder interference ($L_{\text{stick}}(\tau) \geq \Delta z(f) + \delta_{\text{hld}}$).
2) Crib-Constrained Fallback Synthesis: When an exact twist drill for diameter D_h is absent from $\mathcal{C}$, TAMP automatically synthesizes either (i) helical bore interpolation using an admissible flat endmill ($D \leq 0.85 D_h$) for medium/large bores, or (ii) pilot drilling with the largest admissible drill $D_{\text{pilot}} < D_h$ in $\mathcal{C}$ (logging explicit setup advisories if a micro-thread cutter such as #2-56 is excluded from the active crib).
3) 10-Stage Technological Precedence DAG: Within each setup ($s \in \{\text{G54}, \text{G55}\}$), operations follow a strict subtractive precedence DAG $\mathcal{G}_{\text{prec}}$ so datum registration and bulk stock removal precede precision finishing and hole/thread operations: (1) T99 WIPS datum probing $\prec$ (2) top/carrier-hat datum facing $\prec$ (3) bulk 2.5D level-set roughing $\prec$ (4) persistent-IPW corner/wall rest-finishing $\prec$ (5) 3D ball-endmill scallop surfacing $\prec$ (6) helical counterbore/bore interpolation $\prec$ (7) 90° spot drilling $\prec$ (8) G83 peck twist drilling $\prec$ (9) countersinking and 45° edge chamfering $\prec$ (10) internal climb thread milling.
4) ATC-Penalized Discrete Tool Selection: Across candidate roughing/finishing/boring tool assignments in $\mathcal{C}$ satisfying $\mathcal{G}_{\text{prec}}$, the discrete planner minimizes estimated shop time including physical automatic-tool-changer swap latency $t_{\text{ATC}} = 6.0$ s:

$$J_{\text{task}}(\mathcal{S}) = \sum_{o \in \mathcal{S}} \frac{\widehat{V}_o}{\text{MRR}(m_{\text{stock}}, \tau_o)} + \big(|\mathcal{U}(\mathcal{S})| - 1\big)\, t_{\text{ATC}}, \tag{3}$$

where $\mathcal{U}(\mathcal{S})$ is the set of unique tools across sequence $\mathcal{S} = (o_1, \ldots, o_O)$ and $\widehat{V}_o$ is the dexel volume assigned to operation o. Equation (3) naturally consolidates operations onto shared cutters (e.g., reusing a single $1/4''$ flat endmill for both 2.5D rest-finishing and helical counterboring) whenever an extra tool change outweighs incremental MRR gains, and assigns persistent magazine numbers T1..Tn across G54 and G55.
5) Chronological Persistent-IPW Propagation: Rather than planning each operation against an unmachined block, the exact 2.5D dexel heightfield carved by optimized operation o becomes the initial stock state for operation $o + 1$:

$$z_{\text{IPW}}^{(o+1)}(x, y) \leftarrow z_{\text{final}}^{(o)}(x, y), \qquad o = 1, \ldots, O-1, \tag{4}$$

with $z_{\text{final}}^{(O_1)}$ inverted via $\mathcal{F}$ (Eq. (2)) at the G54→G55 boundary. Downstream rest-finishing, 3D ball-scallop, and hole/thread operations thus sense exact remaining stock, enter cuts just above the local IPW ceiling, and convert zero-removal traverses into collision-checked rapid bridges.

C. Clean-Room feeds_and_speeds Physics Engine

Every discrete TAMP evaluation and continuous CWE optimization queries our clean-room physics module (cam.feeds_and_speeds). Each alloy m_{stock} specifies Brinell hardness, specific cutting energy k_p (N/mm^2, e.g., 700 for Al 6061-T6, 850 for 7075-T6, 2100 for 1018 Steel,

3400 for Ti-6Al-4V), reference carbide surface speed $V_{c,0}$, reference chip load $f_{z,0}(D)$, and slotting derating γ_{slot}. Given cutter substrate ($E = 600$ GPa carbide, 210 GPa HSS/cobalt), diameter D, flute count z_f, stickout L_{stick}, axial depth a_p, and radial immersion a_e:

1) Spindle Speed and Thinning-Compensated Target Chip Load: Spindle speed is clamped to machine bounds via $N_{\text{RPM}} = \text{clip}\big(\frac{1000\,V_c}{\pi D}, N_{\min}, N_{\max}\big)$ with tooth-passing frequency $f_{\text{tooth}} = (N_{\text{RPM}}/60)\,z_f$ and cutting speed $v_c = \pi D N_{\text{RPM}}/60000$ m/s. Target undeformed chip thickness is $h_{\text{opt}} = \gamma_{\text{op}}\, f_{z,0}(D)$ (scaled by γ_{slot} when $a_e \geq 0.85D$). Under partial radial immersion ($u = a_e/D < 0.5$) or shallow ball-endmill axial engagement ($a_p < R$), nominal feed per tooth boosts by $\min\big(2.2, \psi(u)^{-1}\big)$ (and $\min\big(2.0, \sqrt{R/\max(a_p, \varepsilon)}\big)$ for ball mills) so actual chip thickness matches h_{opt}.
2) Six Simultaneous Machine and Structural Clamps: With volumetric removal rate $Q(f) = a_p a_e f$ (mm^3/s), cutting power $P(f) = 10^{-3} k_p Q(f)$ (W), tangential cutting force $F_c(f) = P(f)/v_c$ (N), and effective flute-core second moment of area $I = \frac{\pi}{64}(0.62D)^4$, linear feedrate f (mm/s) is simultaneously bounded by machine feed, spindle power, spindle torque $\tau_{\text{sp}}(N_{\text{RPM}})$, tool-core shear torque τ_{core}, frame force $F_{\max}$, and cantilever tip deflection $\delta = \frac{F_c L_{\text{stick}}^3}{3EI} \leq \delta_{\max}$ (0.012 mm finishing, 0.035 mm roughing):

$$f_{\text{lim}} = \min\Big(f_{\max},\ \frac{10^3 \eta P_{\max}}{k_p a_p a_e},\ \frac{10^3 \tau_{\text{sp}} \omega}{k_p a_p a_e},\ \frac{10^3 \tau_{\text{core}} \omega}{k_p a_p a_e},\ \frac{10^3 F_{\max} v_c}{k_p a_p a_e},\ \frac{3 \cdot 10^3 EI \delta_{\max} v_c}{k_p a_p a_e L_{\text{stick}}^3}\Big), \tag{5}$$

where $\omega = 2\pi N_{\text{RPM}}/60$. For twist drilling, axial thrust and torque determine feed per revolution f_{rev} and automatically switch from continuous G81 drilling to G83 chip-breaking peck cycles ($\Delta z_{\text{peck}} = \kappa_{\text{peck}} D$) whenever $L_h/D > 2.2$.

D. Clean-Room CAM Seed Trajectory Synthesis

Each discrete operation $o \in \mathcal{S}$ initializes a collision-free baseline trajectory $\bar{\boldsymbol{\tau}}^{(o)}$ via one of six clean-room geometric generators (cam.strategies, cam.arbitrary_cad_ops):

- Datum Facing & Boss Contouring: Generates bidirectional zig-zag raster passes with lead-in/out extensions beyond stock XY bounds ($a_e = 0.68D$) for top/hat facing, and exterior offset perimeter passes for boss profiling.
- Euclidean-EDT 2.5D Level-Set & Persistent-IPW Rest Clearing: For arbitrary non-convex STEP/STL pockets, islands, and ribs at axial slice z_ℓ, we rasterize the forbidden part mask $\mathcal{O}(z_\ell) = \{(x, y) : z_{\text{tgt}}(x, y) > z_\ell + \varepsilon_z\}$ at pitch Δx and compute the exact 2D Euclidean Distance Transform $d_{\text{EDT}}(x, y)$ outside $\mathcal{O}(z_\ell)$. To guarantee zero chordal facet gouging on curved boundaries, minimum cutter-center clearance is set to $d_0 = R + \delta_{\text{stock}} + 0.55\Delta x$. Sub-cell Marching Squares extracts nested iso-contours $\{d_{\text{EDT}} = d_0 + m\,a_e\}_{m \geq 0}$ ordered inside-out (or outside-in from open billet edges). For rest-finishing, contours are pruned to active regions where persistent IPW exceeds slice depth ($z_{\text{IPW}}(x, y) - z_\ell > \varepsilon_z$). Enclosed contours enter via a 2.5° helical ramp starting 0.5 mm above the local IPW ceiling $\max_{\mathcal{N}} z_{\text{IPW}}$, and non-adjacent loops link at verified local IPW clearance.
- Trochoidal Adaptive Channel Clearing: Advances a guiding medial centerline by pitch $p_{\text{troch}} \leq 0.22D$ superimposed with circular trochoidal loops so instantaneous radial immersion remains bounded in narrow slots [16].
- Inverse-Offset 3D Ball-Endmill Scallop Surfacing: Given a 3D freeform B-spline region Ω_{3D} and a ball endmill of radius $R_b = D/2$, parallel finishing stepover is chosen from target cusp height h_{scal} via $s = \sqrt{8 R_b h_{\text{scal}}}$. To guarantee exact gouge-free contact across steep walls and saddle curvature [3], the cutter-tip height $z_{\text{CL}}(x, y)$ is evaluated via the exact inverse-offset envelope over disk $\xi^2 + \eta^2 \leq R_b^2$:

$$z_{\text{CL}}(x, y) = \max_{\xi^2+\eta^2 \leq R_b^2} \Big[z_{\text{tgt}}(x + \xi, y + \eta) + \sqrt{R_b^2 - \xi^2 - \eta^2} - R_b \Big], \tag{6}$$

clipped strictly to the eroded interior of Ω_{3D} and clamped below by the surrounding pocket floor $z_{\text{floor}}(\Omega_{\text{3D}})$ so the spherical flank never undercuts adjacent planar shelves.
- Helical Bore Interpolation, G83 Drilling, Countersinking, & 45° Chamfering: Counterbores and flat-bottom bores interpolate $(x_h + r_b \cos\theta,\, y_h + r_b \sin\theta,\, z_0 - \frac{p_z}{2\pi}\theta)$ at radius $r_b = \frac{1}{2}(D_{\text{bore}} - D)$ and helical pitch $p_z \leq \min(0.25D, 1.2\text{ mm})$ followed by a 360° flat cleanup pass; spot drills and countersinks dwell briefly at exact cone depth; G83 peck cycles expand into explicit plunge/retract/re-entry segments for exact simulator timing; and 90° chamfer mills offset laterally by Δr_{ch} and axially by $-\big(\Delta r_{\text{ch}} + w_{\text{ch}}\big)$ so the 45° bevel engages away from the dead tool tip.
- Internal Climb Helical Thread Milling: Right-hand internal threads start at hole center at z_{bot}, arc tangentially onto major-thread radius $r_{\text{thr}} = \frac{1}{2}(D_{\text{major}} - D_{\text{tool}})$, climb-mill upward $\big(x_h + r_{\text{thr}} \cos\theta,\, y_h + r_{\text{thr}} \sin\theta,\, z_{\text{bot}} + \frac{P}{2\pi}\theta\big)$ synchronized to exact thread pitch P, and arc back to center before retracting.

IV. Differentiable Cutting Simulation, Blended Kinematics, and Co-Optimization

A. 3D Swept-Cylinder SDF and Smooth-to-Hard STE Continuation

For full 3D volumetric verification and first-order policy/trajectory benchmarking, the stock billet is repre-

sented on a 3D Eulerian grid $\Omega \subset \mathbb{R}^3$ of pitch Δx with signed distance field $\phi_t : \Omega \to \mathbb{R}$ after segment $t \in \{0, \dots, T\}$ ($\phi_t(\mathbf{x}) < 0$ inside solid metal). For a cutter of radius R, flute length L_{flute}, and holder radius $R_{\text{holder}} > R$ sweeping from $\mathbf{p}_{t-1}$ to $\mathbf{p}_t$ ($\Delta\mathbf{p}_t = \mathbf{p}_t - \mathbf{p}_{t-1}$), let $\tau^\star(\mathbf{x}) = \text{clip}\big(\frac{(\mathbf{x}-\mathbf{p}_{t-1})\cdot\Delta\mathbf{p}_t}{\|\Delta\mathbf{p}_t\|_2^2+\varepsilon}, 0, 1\big)$ and $\mathbf{c}_t(\mathbf{x}) = \mathbf{p}_{t-1} + \tau^\star(\mathbf{x})\Delta\mathbf{p}_t$. The swept flute SDF is

$$\psi_{\text{tool}}(\mathbf{x}; \mathbf{p}_{t-1}, \mathbf{p}_t) = \max\Big(\|\mathbf{x}^{xy} - \mathbf{c}_t^{xy}(\mathbf{x})\|_2 - R, \; c_t^z(\mathbf{x}) - x^z, \; x^z - \big(c_t^z(\mathbf{x}) + L_{\text{flute}}\big)\Big). \tag{7}$$

Hard Boolean subtraction updates $\phi_t^{\text{hard}}(\mathbf{x}) = \max\big(\phi_{t-1}^{\text{hard}}(\mathbf{x}), -\psi_{\text{tool}}(\mathbf{x})\big)$. To obtain dense spatial derivatives across stock boundaries without low-stiffness volumetric bias, we pair exact Boolean evaluation with a stabilized log-sum-exp surrogate at stiffness $k > 0$:

$$\phi_t^{(k)}(\mathbf{x}) = m + \frac{1}{k}\log\Big(e^{k(\phi_{t-1}^{(k)}(\mathbf{x})-m)} + e^{k(-\psi_{\text{tool}}(\mathbf{x})-m)}\Big), \tag{8}$$

where $m = \max(\phi_{t-1}^{(k)}(\mathbf{x}), -\psi_{\text{tool}}(\mathbf{x}))$ and occupancy is $\sigma_k(\phi) = (1 + e^{k\phi})^{-1}$. Across step $n \in \{0, \dots, N-1\}$, we anneal $k(n)$ linearly from $k_{\min} = 25$ to $k_{\max} = 90$ inside a Straight-Through Estimator (STE) [2]:

$$\mathcal{L}_{\text{STE}}(\boldsymbol{\theta}; k(n)) = \mathcal{L}_{\text{hard}}(\boldsymbol{\theta}) + \Big(\mathcal{L}_{k(n)}(\boldsymbol{\theta}) - \text{sg}\big[\mathcal{L}_{k(n)}(\boldsymbol{\theta})\big]\Big), \tag{9}$$

so model selection and feasibility checks use exact Boolean $\mathcal{L}_{\text{hard}}$, while reverse-mode Taichi kernels propagate $\nabla_{\boldsymbol{\theta}}\mathcal{L}_{k(n)}$.

B. Blended CNC Kinematics and Exact Junction-Velocity Gradients

Modern CNC lookahead interpolators avoid full stops (G61) at gentle G1 junctions by blending velocity vectors within path tolerance (RS274 G64 P0.05, Haas G187 P2 E0.05) [5, 13]. Given segment chord $\ell_k = \|\Delta\mathbf{p}_k\|_2$, unit tangent $\hat{\mathbf{d}}_k = \Delta\mathbf{p}_k/\ell_k$, commanded speed v_k (f_k on G1, v_{rapid} on G0), and velocity vector $\mathbf{v}_k = v_k\hat{\mathbf{d}}_k$, acceleration-bounded transition duration over axis limit $a_{\max}$ yields

$$\tau_k(\mathbf{p}, \mathbf{f}) = \frac{\sqrt{\ell_k^2 + \varepsilon_L^2} - \varepsilon_L}{v_k} + \frac{\sqrt{\|\mathbf{v}_k - \mathbf{v}_{k-1}\|_2^2 + \varepsilon_v^2} - \varepsilon_v}{2\,a_{\max}}, \tag{10}$$

plus terminal stop $\|\mathbf{v}_M\|_2/(2a_{\max})$ and total time $T_{\text{cycle}} = \sum_{k=1}^{M}\tau_k$.

A crucial subtlety arises on dense Marching-Squares and 3D surfacing polylines with hundreds of short chords ($\ell_k \sim 0.5$–2 mm): boosting v_k while ignoring the second term of Eq. (10) inflates $\|\mathbf{v}_k - \mathbf{v}_{k-1}\|_2/(2a_{\max})$ at every corner turn, increasing true cycle time! Differentiating T_{cycle} exactly with respect to each station speed v_k couples every segment to both its predecessor $k-1$ and successor $k+1$:

$$\begin{aligned}\frac{\partial T_{\text{cycle}}}{\partial v_k} = &-\frac{\sqrt{\ell_k^2 + \varepsilon_L^2} - \varepsilon_L}{v_k^2} \\ &+ \frac{(\mathbf{v}_k - \mathbf{v}_{k-1})\cdot\hat{\mathbf{d}}_k}{2\,a_{\max}\sqrt{\|\mathbf{v}_k - \mathbf{v}_{k-1}\|_2^2 + \varepsilon_v^2}} \\ &+ \frac{(\mathbf{v}_k - \mathbf{v}_{k+1})\cdot\hat{\mathbf{d}}_k}{2\,a_{\max}\sqrt{\|\mathbf{v}_{k+1} - \mathbf{v}_k\|_2^2 + \varepsilon_v^2}}.\end{aligned} \tag{11}$$

For collinear segments ($\hat{\mathbf{d}}_{k-1} = \hat{\mathbf{d}}_k = \hat{\mathbf{d}}_{k+1}$), telescopic cancellation leaves pure traversal speedup $-\ell_k/v_k^2$; at sharp turns ($\hat{\mathbf{d}}_k \cdot \hat{\mathbf{d}}_{k+1} < 1$), positive junction sensitivity automatically decelerates v_k before the corner when ℓ_k is short relative to $v_k^2/a_{\max}$ while sprinting on straightaways.

C. Chronological 2.5D Dexel CWE, Chip Thinning, and Air Shortcutting

To optimize directly for constant undeformed chip thickness and minimal cycle time (SubsegmentedCWEModel and optimize_arbitrary_cwe_plan), long G1 moves are subdivided to ≤ 2.0 mm stations ($k = 1, \dots, M$) and swept chronologically over the 0.25 mm 2.5D dexel grid z_{stock}:

1) Endcap-Corrected Radial Immersion and Martellotti Chip Thinning: Because sweeping a disk of diameter D along a short chord ℓ_k^{xy} carves both a rectangular swath and circular endcaps, dividing removed volume ΔV_k by ℓ_k^{xy} alone overestimates radial width on short segments. Normalizing by effective swept length $\ell_k^{\text{eff}} = \ell_k^{xy} + \kappa_{\text{cap}} D$ ($\kappa_{\text{cap}} = 0.60$) yields $a_{e,k} = \text{clip}\big(\frac{\Delta V_k}{a_{p,k}\,\ell_k^{\text{eff}}}, 0, D\big)$ and fractional immersion $u_k = a_{e,k}/D$. Actual undeformed chip thickness [14] is then

$$\begin{aligned}\psi(u_k) &= \begin{cases}\sqrt{4u_k(1-u_k)}, & 0 < u_k < 0.5, \\ 1, & u_k \geq 0.5,\end{cases} \\ h_{\text{ex},k} &= \frac{f_k}{f_{\text{tooth}}}\,\psi(u_k).\end{aligned} \tag{12}$$

Forward evaluation uses exact hard dexel state; reverse-mode STE spatial gradients $\partial\Delta V_k/\partial\mathbf{p}$ and analytical junction/chipload gradients $\partial\mathcal{L}_{\text{CWE}}/\partial f_k$ adjust lateral engagements and per-station feeds so $h_{\text{ex},k} \to h_{\text{opt}}$ on engaged cuts while zero-engagement air moves accelerate up to $f_{\max}$.

2) Full-Radius-Verified Empty-Air Shortcutting: After continuous STE-Adam convergence, contiguous runs of horizontal G1 stations verified against the chronological dexel IPW to remove $\Delta V_k = 0$ are tested for shortcut replacement: if sweeping the full physical cutter envelope ($R + 0.06$ mm) along a direct chord or lifted G0 clearance bridge clears all current z_{stock} and target walls, the redundant air loop is collapsed into a fast bridge—yielding substantial savings even when cutting feeds are held strictly identical across methods.

For 3D SDF benchmarking (Exp1–Exp4), the composite objective combines normalized cycle time, target gouge/residual, Peak MRR barrier $\mathcal{B}_{\mathrm{MRR}} = w_{\mathrm{MRR}} \sum_{t=1}^{T} \max(0, \mathrm{MRR}_t/\mathrm{MRR}_{\max} - 1)^2$, 2D wall-clearance $\mathcal{B}_{\mathrm{wall}}$, cylindrical holder $\mathcal{B}_{\mathrm{holder}}$, and rapid-stock $\mathcal{B}_{\mathrm{rapid}}$ barriers:

$$\begin{aligned}\mathcal{L}_{\mathrm{3D}} = w_T \frac{T_{\mathrm{cycle}}}{T_{\mathrm{nom}}} + w_g \bar{V}^2_{\mathrm{gouge}} + w_r \bar{V}^2_{\mathrm{res}} \\ + \mathcal{B}_{\mathrm{MRR}} + \mathcal{B}_{\mathrm{wall}} + \mathcal{B}_{\mathrm{holder}} + \mathcal{B}_{\mathrm{rapid}},\end{aligned} \tag{13}$$

while CWE co-optimization (Exp5–Exp7) minimizes $\mathcal{L}_{\mathrm{CWE}} = \frac{T_{\mathrm{cycle}}}{T_{\mathrm{nom}}} + w_h \sum_{k\in\mathcal{E}} \frac{\tau_k}{T_{\mathrm{nom}}} \left(\frac{h_{\mathrm{ex},k} - h_{\mathrm{eff},k}}{h_{\mathrm{opt}}}\right)^2 + \mathcal{B}_{\mathrm{MRR}} + \mathcal{B}_{\mathrm{res}} + \mathcal{B}_{\mathrm{smooth}}$.

D. Drift-Free Parameterizations, RL Policies, Fast Overfitting, and 6-Dialect Post

Wall-Pinned Drift-Free Parameterization: Perturbing relative steps $\Delta\mathbf{p}_t$ causes $O(T)$ random-walk drift. Instead, we pin axial heights $\bar{p}_i^z$ and bound horizontal waypoint offsets by exact analytical Euclidean wall clearance $d_{\mathrm{wall}}(\bar{\mathbf{p}}_i)$:

$$\begin{aligned}\mathbf{p}_i^{xy}(\boldsymbol{\theta}) &= \bar{\mathbf{p}}_i^{xy} + r_i \tanh(\mathbf{u}_{xy,i}), \\ r_i &= \mathrm{clip}\big(d_{\mathrm{wall}}(\bar{\mathbf{p}}_i) - \delta_{\mathrm{pin}},\, 0,\, r_{\max}\big),\end{aligned} \tag{14}$$

using either clamped cubic B-splines $\mathbf{u}_{xy} = \mathbf{B}\boldsymbol{\theta}_{\mathrm{pos}}$ ($K_p \le 40$, $r_{\max} = 1.25$ mm) in 3D or vertical-column waypoint groups ($r_{\max} = 3.0$ mm) keeping vertical plunges strictly vertical in 2.5D CWE, paired with sigmoid feeds $f_k \in [f_{\min}, 0.995\, v_{\max}]$.

Generalized Residual RL vs. Fast Inference-Time Trajectory Overfitting: DiffCAM supports three deployment modes:

1) Direct Per-Part Differentiable Optimization (STE-Adam / DiffCAM-Chipload): Optimizes $\boldsymbol{\theta} \in \mathbb{R}^D$ ($D \approx 250$–650) directly on the input part's CAM seed with zero neural training.
2) Amortized Offline RL (DiffPolicy-SHAC / Hard-PPO): Trains a compact two-layer 64–64 tanh MLP $\boldsymbol{\pi}_\psi(\mathbf{a}_k \mid \mathbf{s}_k)$ offline on development parts—taking local chord length, turn angle, normalized wall clearance, nominal engagement u_k, and alloy/tool context as state $\mathbf{s}_k$—via analytical backprop-through-physics (SHAC [24]) or clipped policy gradients with GAE (PPO [18, 19]), inferring $\boldsymbol{\theta}_0 = \boldsymbol{\pi}_{\psi^\star}(\mathbf{S}(\mathcal{P}_{\mathrm{test}}))$ in $<$3.5 ms.
3) Hybrid Warm-Start Inference-Time Overfitting (SHAC + DiffCAM-Overfit-K): Detaches the MLP at test time, sets $\boldsymbol{\theta}^{(0)} \leftarrow \boldsymbol{\pi}_{\psi^\star}(\mathbf{S}(\mathcal{P}_{\mathrm{test}}))$, and takes $K \in \{4, \ldots, 32\}$ analytical CWE STE-Adam updates directly in $\mathbb{R}^D$ plus air shortcutting—eliminating policy approximation error in sub-second wall time where zero-order RL fine-tuning stalls under $\mathcal{O}(D/\sigma^2)$ variance.

6-Dialect Multi-Setup Post-Processing and 3D Tool-Library WebGL Simulator: Clean-room post-processors (cam.post_processors) compile multi-setup programs across 6 industrial controllers (Haas NGC, LinuxCNC RS274, FANUC 31i, FANUC Robodrill, Makino Pro 5, and Mazak SmoothG ISO), emitting safe start blocks, optional T99 G65 P9995/P9811/P9814 Renishaw WIPS touch-probing headers, Tn M6 / G43 Hn tool-length compensation, G187/G05.1 smoothing codes, M00 Op1→Op2 flip stops (G54→G55), and exact bi-directional segment-to-line maps. Our web interface (scripts/serve_simulator.py, viewer/) allows users to upload arbitrary .step/.stl parts, switch between US/Metric/Custom Tool Cribs, tune aggressiveness and alloys, and inspect 3-way side-by-side stock carving with realistic 3D holder + flute/drill/ball/chamfer/thread-mill assemblies synchronized line-by-line with live 6-dialect G-code.

V. Experimental Setup, Split Protocol, and Metrics

Benchmark Suites: (1) Single-Tool 16-Task Suite (Exp1–Exp5b): Four clean-room 2.5D CAM strategy families—facing_open_boss (37–39 segs), pocket_island (92–169 segs), adaptive_channel (261–370 segs), and contour_profile (156–188 segs)—with 2 dev and 2 held-out test parts per family (16 tasks, Al 6061-T6 billets 30–38 × 24–32 × 8 mm, ∅5.0 mm 3-flute carbide endmill at 8000 RPM, $v_{\max} = 20$ mm/s, $v_{\mathrm{rapid}} = 100$ mm/s, $a_{\max} = 150$ mm/s^2). (2) Multi-Tool .stl Suite (Exp6): Four multi-feature 3D binary .stl solids across Al 6061-T6, Al 7075-T6, and Mild Steel 1018. (3) Arbitrary ISO 10303-21 .step Suite (Exp7): All 10 parts of the industry-standard TITAN-1M through TITAN-10M CNC building-block series (example_parts/Titan-M1.step..M10.step, authored in inches with 49–203 mm spans, multi-pocket webs, radiating ribs, blind/through holes, counterbores, 82° countersinks, 45° chamfers, ANSI Unified internal threads, underside cavities requiring G54+G55 flips, and 3D B-spline domes/airfoils), machined in Al 6061-T6 on a Haas VF-2/Mini-Mill envelope (10,000 RPM, 15 kW, $a_{\max} = 1500$ mm/s^2).

Role of dev vs. Held-Out test Splits: Direct per-part optimizers (STE-Continuation-Adam, DiffCAM-Chipload, CEM/CMA-ES) use dev solely to freeze hyperparameters and start cold from each unseen test part's CAM seed. Generalized RL policies (DiffPolicy-SHAC, Hard-PPO) train neural weights $\psi^\star$ strictly offline on dev and evaluate zero-shot ($K = 0$) or with K-step inference-time trajectory overfitting on test. Automated TAMP (Exp6–Exp7) requires zero training split, planning directly from raw .stl/.step geometry.

Physical Evaluation Metrics: We report: (i) Blended Cycle Time T_{cycle} (s) (Eq. (10)), decomposed into engaged cutting ($\Delta V_k > 0$), unproductive G1 air-cutting ($\Delta V_k = 0$), and G0/ATC swap time (6.0 s/swap); (ii) Fair Same-Speed Time T_{same} vs. Optimal-Chipload Time T_{opt} (s), where T_{same} locks every method to identical baseline cutting feeds f_{base} to isolate pure geometric and

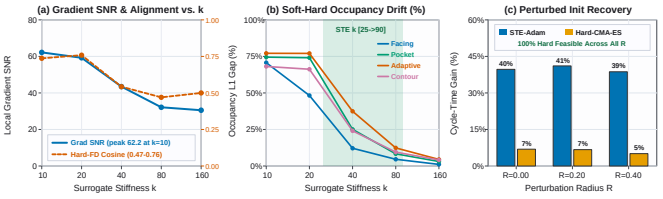


Fig. 3. Exp1: Surrogate gradient conditioning and perturbed recovery. (a) Analytical gradient SNR (62.2 $\rightarrow$ 30.5, blue) and hard finite-difference cosine (0.76 $\rightarrow$ 0.47, orange). (b) Soft-to-hard occupancy L_1 drift collapses from 72.6% ($k = 10$) to 8.7% ($k = 80$). (c) Perturbed initialization recovery ($R \in \{0.0, 0.2, 0.4\}$, 48 calls): STE-Continuation-Adam achieves 38.7%–41.1% gains vs. 5.1%–6.9% for Hard-CMA-ES.

air-shortcut gains; (iii) Target Gouge V_{gouge} and Residual V_{res} (mm^3) plus binary vise-jaw, holder, and rapid-stock collision audits; (iv) Peak MRR (mm^3/s) and Overload (%) above $\text{MRR}_{\max}$; (v) Actual Chip Thickness h_{ex} (mm), RMSE, and Optimal Band (%) inside $[0.56\, h_{\text{opt}}, 1.26\, h_{\text{opt}}]$; and (vi) Gradient SNR, Hard-FD Cosine, L_1 Occupancy Drift, and CAMotics Watertight STL ΔV (%).

VI. Experimental Results and Analysis

A. Exp1: Surrogate Conditioning and Perturbed Reachability

Motivation: Smooth CSG surrogates trade wide gradient support at low stiffness k against Boolean exactness at high k. Across $k \in \{10, 20, 40, 80, 160\}$ (Fig. 3), analytical gradient SNR and cosine similarity to central hard finite differences peak at $k \in \{10, 20\}$ (SNR = **62.19**–**59.24**, $\cos\theta$ = **0.737**–**0.758**) and stay well-conditioned through k = 80–160 (SNR = 32.14–30.53, $\cos\theta$ = 0.47–0.50), while soft-to-hard occupancy L_1 drift drops from **72.6**% ($k = 10$) to **8.7**% ($k = 80$) and **3.2**% ($k = 160$). Annealing $k : 25 \rightarrow 90$ under STE recovers 100% feasibility and 38.7%–41.1% cycle-time savings from perturbed initializations $\|\boldsymbol{\theta}_0\|_\infty \le R \in \{0.0, 0.2, 0.4\}$ (5.8×–7.6× Hard-CMA-ES).

B. Exp2 & Exp3: 16-Task Trajectory vs. RL Benchmark, Transfer, and Ablations

Motivation: Under strict simulator-call parity (B = 64 calls, 3 seeds), how do first-order smooth-to-hard trajectory and RL methods compare against zero-order RL and evolutionary search on exact Boolean cutting? Table I and Fig. 4 show that STE-Continuation-Adam ($-\mathbf{42.0} \pm \mathbf{13.5}$%, 73.91 s) and Annealed-Adam ($-\mathbf{43.0} \pm \mathbf{15.7}$%, 73.74 s) beat fixed-$k$ Naive-Smooth-Adam (−31.9%) by $>$ 10 points and achieve nearly 4× the speedup of Hard-CMA-ES (−11.3%) and Hard-CEM (−9.3%) at 100% feasibility. With identical 64–64 MLPs and 64 calls, first-order DiffPolicy-SHAC ($-\mathbf{37.0} \pm \mathbf{12.7}$%) doubles (2.0×) the speedup of zero-order Hard-PPO ($-\mathbf{18.5} \pm \mathbf{6.6}$%) and leads all methods on enclosed Pocket roughing ($-\mathbf{32.1}$%).

In our 5-arm ablation (Table II bottom, Fig. 5b), freezing feeds without air shortcutting yields 0.2% gain, joint XY-and-feed STE cuts Peak MRR overload by 29% vs. frozen-path feed scheduling (5.2% vs. 7.3%), and removing the MRR barrier ($w_{\text{MRR}} = 0$) triggers $+\mathbf{136.7}$% spindle overload (262.6 mm^3/s). On unseen test transfer (Exp3, Table II top, Fig. 5a), 16 analytical calls cut SHAC overload 4× (16.9% $\rightarrow$ 4.4%).

C. Exp5 & Exp5b: Fair Same-Speed vs. Optimal Chipload and Fast Inference-Time Trajectory Overfitting

Motivation: Exp5 re-evaluates all 8 unseen test parts against a physics-calibrated Nominal-CAM (@2) baseline set to the exact feeds_and_speeds rate (f_{base} = 11.69 mm/s, h_{opt} = 0.0291 mm/tooth), separating Fair Same-Speed (T_{same}) from Optimal-Chipload (T_{opt}) performance (Table III top, Fig. 6a). Exp5b tests whether < 3.5 ms offline RL warm-starting plus $K \in \{4, \ldots, 32\}$ analytical trajectory-overfitting steps can achieve test-part optimality faster than test-time RL (Table III bottom, Fig. 6b).

Three findings stand out: (1) Under Martellotti chip thinning, 3D MRR-only STE-Adam drops to 41.1% optimal-band compliance because capping MRR without $\psi(u_k) < 1$ under-feeds light radial cuts into rubbing. (2) At strictly identical cutting speed f_{base}, DiffCAM-Chipload cuts mean time from 55.21 s $\rightarrow$ **53.60** s (up to $-\mathbf{7.9}$% on adapt-test-01) via corner smoothing and stock-verified G0 air shortcuts, and reaches **48.07** s ($-\mathbf{14.9}$%) with **97.7**% optimal h_{ex} compliance (h_{ex} RMSE 0.0053 mm) under full chipload scheduling. (3) At inference time (Exp5b), PPO-Overfit (160 rollouts, 3.52 s) stagnates at 41.0% optimal h_{ex}, whereas SHAC+Overfit (K = 8–32, **0.48**–**0.93** s) repairs out-of-distribution channels and lifts optimal h_{ex} from 42.4% to **95.6**%–**97.0**%, while Cold-DiffCAM (K = 8, **0.59** s) hits **98.2**% optimal h_{ex} (50.16 s).

D. Exp4: Multi-Controller NC and CAMotics Watertight STL Audit

Table IV certifies all 8 held-out test parts compiled across 6 CNC dialects (48/48 pass, 0.00 mm^3 gouge/residual at 0.125 mm step), with containerized CAMotics v1.2.0 reconstructing 8/8 closed watertight STLs matching nominal enclosed volume to ΔV = **0.000**%.

E. Exp6: Automated Multi-Tool .stl TAMP Across Three Alloys

Figure 7 and Table V evaluate single-setup multi-tool TAMP across four .stl parts and three alloys (Al 6061-T6, Al 7075-T6, Mild Steel 1018). TAMP automatically sizes billets and vise jaws, reuses T_1 (⌀10 mm) for bulk roughing where ATC-amortized (0 s extra swap), derates spindle RPM in 1018 Steel (4024–4713 RPM vs. 8000 RPM in 6061), and propagates persistent IPW so T_2 rest-finishing shortcuts cleared interiors—reducing cutting time by up

TABLE I

Exp2 Main Benchmark across 16 Milling Tasks (4 Families × 4 Geometries × 3 Seeds, Strict Budget $B = 64$ Simulator Calls). Cycle-time reductions ($-\Delta T_{\text{cycle}}$, %) are mean ± std relative to 10 mm/s Nominal-CAM; all methods achieve 100% hard Boolean feasibility under blended kinematics.

Method	Optimization Class	Facing (%)	Pocket (%)	Adaptive (%)	Contour (%)	Overall $-\Delta T$ (%)	Mean T (s)	Peak MRR (mm^3/s)
Nominal-CAM	Clean-Room CAM Seed	0.0 ± 0.0	0.0 ± 0.0	0.0 ± 0.0	0.0 ± 0.0	0.0 ± 0.0	122.20	75.4
Hard-CEM [17]	0th-Order Elite Search	15.6 ± 2.6	6.3 ± 3.4	6.1 ± 1.2	9.3 ± 2.0	9.3 ± 4.5	112.17	88.8
Hard-CMA-ES [8]	0th-Order Covariance ES	22.3 ± 2.1	6.3 ± 3.5	6.0 ± 1.8	10.6 ± 3.4	11.3 ± 7.2	110.62	95.9
Hard-PPO [19]	0th-Order Clipped RL	21.0 ± 3.4	20.5 ± 5.6	10.1 ± 4.4	22.2 ± 4.5	18.5 ± 6.6	102.26	93.9
Naive-Smooth-Adam	1st-Order Fixed $k = 25$	33.6 ± 8.1	24.2 ± 14.0	33.3 ± 4.4	36.4 ± 5.2	31.9 ± 9.9	83.44	105.9
DiffPolicy-SHAC [24]	1st-Order Analytic RL	53.6 ± 2.8	$\mathbf{32.1 \pm 12.4}$	24.6 ± 4.2	37.7 ± 2.6	37.0 ± 12.7	81.67	107.4
Annealed-Adam	1st-Order 4:1 k-Sched	57.4 ± 3.1	29.2 ± 19.2	33.0 ± 3.5	$\mathbf{52.4 \pm 3.4}$	$\mathbf{43.0 \pm 15.7}$	**73.74**	109.7
STE-Continuation-Adam	1st-Order 1:1 STE (Ours)	$\mathbf{59.7 \pm 1.2}$	26.9 ± 9.7	$\mathbf{37.3 \pm 7.2}$	44.1 ± 3.7	$\mathbf{42.0 \pm 13.5}$	73.91	113.3

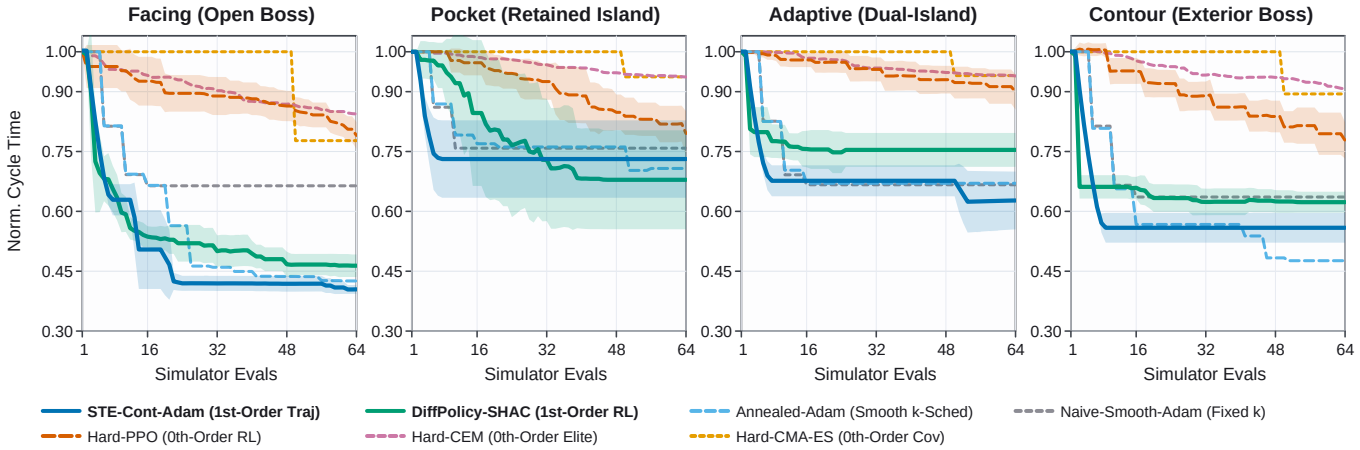


Fig. 4. Exp2 sample-efficiency convergence across clean-room CAM families ($B = 64$ simulator calls, mean ± 1 SD). First-order methods (STE-Continuation-Adam, Annealed-Adam, DiffPolicy-SHAC) achieve steep reductions within 16–24 calls via exact per-segment sensitivities, whereas zero-order RL and ES progress slowly over 100–370 segments.

TABLE II

Top: Exp3 3D Multi-Task Policy Transfer (8 Dev → 8 Unseen Test). Bottom: Exp2 5-Arm Physical & Kinematic Ablations (8 Test, $B = 64$).

Regime / Ablation Arm	Calls	$-\Delta T_{\text{cycle}}$ (%)	Peak MRR	Overload
Exp3: Unseen Test Policy Generalization (3D MRR Surrogate)				
Hard-PPO-ZeroShot	1	26.1 ± 9.5	133.2	24.5%
DiffPolicy-SHAC-ZeroShot	1	8.3 ± 17.9	124.9	16.9%
Hard-PPO-FewShot16	16	23.4 ± 9.5	104.9	2.7%
DiffPolicy-SHAC-FewShot16	16	$\mathbf{27.7 \pm 15.8}$	111.4	4.4%
Scratch-STE-Adam-16	16	$\mathbf{40.1 \pm 7.4}$	114.4	4.0%
Exp2-Abl: Physical & Kinematic Ablations ($B = 64$)				
Path-Only-Frozen-Feed	64	0.2 ± 0.4	82.1	0.0%
Full-Joint-Blended-STE	64	37.0 ± 17.9	114.9	**5.2**%
Exact-Stop-Surrogate	64	39.2 ± 18.2	109.6	2.3%
Feed-Only-Frozen-Path	64	$\mathbf{39.5 \pm 13.7}$	118.7	7.3%
No-MRR-Barrier ($w_{\text{MRR}} = 0$)	64	66.2 ± 3.5	**262.6**	**136.7**%

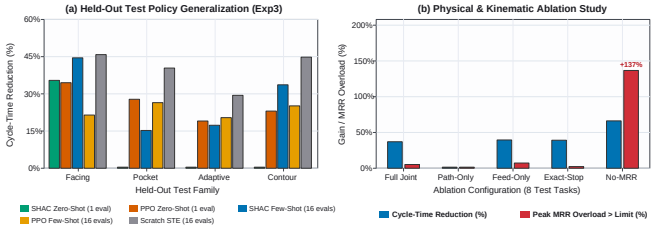


Fig. 5. Exp3 policy generalization and physical ablations. (a) Held-out test cycle-time reductions across families for zero-shot (1 call) and few-shot (16 calls) transfer. (b) Ablating feed optimization collapses gains to 0.2%; ablating the MRR barrier triggers +136.7% cutter overload.

to **43.7**% (**36.7**% total incl. 18 s ATC) with **4/4** safety pass (0.00 mm^3 gouge).

F. Exp7: Arbitrary ISO 10303-21 STEP Benchmark on TITAN-1M–10M

Motivation: To test true "slicer-like" end-to-end generality on real-world multi-setup, freeform, and threaded engineering CAD parts, Exp7 evaluates DiffCAM directly on all 10 unaltered ISO 10303-21 .step files of the TITAN-1M–TITAN-10M aerospace building-block curriculum (Titan-M1.step–Titan-M10.step) constrained to our US Imperial Shop Tool Crib.

Figure 8 and Table VI summarize all 10 parts (17.0 s mean end-to-end STEP parsing, setup scoring, TAMP sequencing, and CWE optimization wall time per part):

- Automated 6-Axis $SO(3)$ Reorientation & Multi-Setup Soft-Jaw Flipping: Six parts (M2, M3, M5, M6, M9, M10) authored with primary features along $+Y$ are automatically rotated $+Y \mapsto +Z$ by Eq. (1). Four parts (M4, M7, M8, M10) with top-occluded underside cavities or full-depth profiled silhouettes automatically synthesize two-setup Op1 (G54) → M00 → Op2 (G55) plans, facing off the sacrificial carrier hat and finishing underside pockets/counterbores

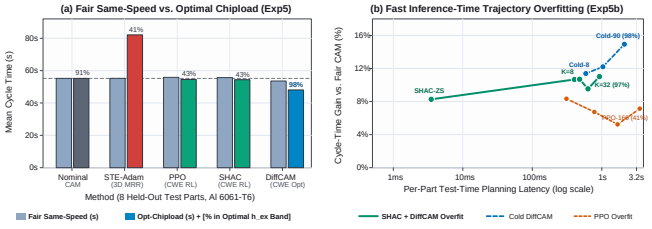


Fig. 6. Exp5 fair chipload comparison and Exp5b inference-time trajectory overfitting. (a) Mean Same-Speed (T_{same}) and Optimal-Chipload (T_{opt}) cycle times across 8 unseen test parts, annotated with % of engaged cutting inside the optimal h_{ex} band. (b) Test-time latency vs. cycle-time gain: warm-starting from SHAC-ZeroShot and running $K \in \{4, 8, 16, 32\}$ analytical CWE trajectory-overfitting steps rapidly dominates zero-order PPO-Overfit (16–160 rollouts) and converges cleanly alongside cold-start DiffCAM.

TABLE III
Top (Exp5): Fair Same-Speed (T_{same}) vs. Optimal-Chipload (T_{opt}) CWE Comparison across 8 Unseen Test Parts (Al 6061-T6, $f_{\text{base}} = 11.7$ mm/s). Bottom (Exp5b): Fast Inference-Time Trajectory Overfitting vs. Test-Time RL.

Method / Test-Time Regime	Evals	Latency	T_{same} (s)	T_{opt} (s)	$-\Delta T_{\text{opt}}$	Opt h_{ex}
Exp5: Fair Baseline vs. Chipload Optimization (8 Test Parts, 100% Posted Pass)						
Nominal-CAM (@2)	0	—	55.21	55.21	0.0%	90.9%
STE-Adam (3D MRR)	64	1.6 s	55.24	82.12	−38.7%	41.1%
PPO-Chipload (RL)	0†	2.1 ms	55.86	54.57	+9.3%	42.6%
SHAC-Chipload (RL)	0†	3.5 ms	55.71	54.41	+9.5%	42.7%
DiffCAM-Chipload ($K=90$)	90	2.12 s	**53.60**	**48.07**	**+14.9%**	**97.7%**
Exp5b: Fast Test-Time Trajectory Overfitting vs. RL (†trained offline on dev)						
PPO-Overfit ($K=16$)	16	0.78 s	56.02	56.85	+6.7%	40.5%
PPO-Overfit ($K=160$)	160	3.52 s	56.03	56.25	+7.1%	41.0%
SHAC+Overfit ($K=4$)	4	0.40 s	56.70	52.27	+10.7%	62.4%
SHAC+Overfit ($K=8$)	8	0.48 s	56.12	51.29	+10.7%	80.7%
SHAC+Overfit ($K=16$)	16	0.63 s	55.20	51.14	+9.5%	95.6%
SHAC+Overfit ($K=32$)	32	**0.93** s	55.19	50.26	+11.0%	97.0%
Cold-DiffCAM ($K=8$)	8	**0.59** s	55.15	50.16	+11.4%	98.2%
Cold-DiffCAM ($K=32$)	32	**1.04** s	**54.45**	**49.58**	**+12.2%**	**98.5%**

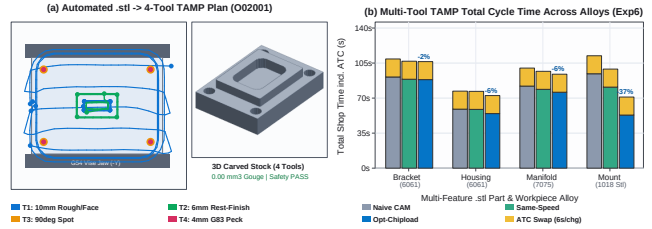


Fig. 7. Exp6 automated multi-tool .stl TAMP. (a) 4-tool G54 plan and carved stock on tamp-flanged-bracket (O02001, 0.00 mm^3 gouge). (b) Total shop cycle time (incl. 6 s ATC swaps) across four parts and three alloys.

in flipped coordinates (Eq. (2)) with zero vise-jaw collisions.

- Multi-Class Crib Sequencing, Thread Milling, & 3D Ball Scallop: Across the 10 parts, TAMP sequences 6–11 unique US tools (7–13 operations, 4,743–19,927 Haas NGC lines/part), combining 2″ face mills, 5/8″ roughers, 1/4″ rest/helical-bore flat mills, 1/4″ ball endmills for gouge-free inverse-offset 3D B-spline surfacing on Titan-M9 (6 dome regions) and Titan-M10 (8 airfoil regions), 90° spot drills, wire/fractional G83 drills, 82° countersinks, 45° chamfer mills, and internal climb thread mills (M3, M4, M5, M10). On Titan-M1, when an exact ⌀2.06 mm drill and #2-56 micro-thread mill are absent from the standard crib, TAMP safely falls back to a #47 (0.0785″) wire pilot drill and emits a clear crib warning.
- Consistent Gains from Exact G187 Junction Gradients & Zero Gouge: Every single part (10/10) satisfies $T_{\text{opt}} < T_{\text{same}} < T_{\text{naive}}$ with **0.000** mm^3 target gouge and 0 vise, holder, or rapid collisions. Fair matched-speed air shortcutting reduces mean cycle time from 2363.2 s → **1488.8** s (**−36.3**%), while joint CWE + exact G187 junction-velocity optimization (Eq. (11)) reaches **1362.5** s (**−41.6**% mean, up to **−54.1**% on 3D-sculptured Titan-M9), compiling cleanly across all 60/60 6-dialect NC programs.

VII. Limitations and Conclusion

Limitations and Remaining Frontiers. While DiffCAM advances end-to-end 3-axis automation from raw ISO 10303-21 .step and .stl CAD solids through 6-sided $SO(3)$ setup orientation, Op1 (G54) / Op2 (G55) soft-jaw flipping, US/Metric crib-constrained tool sequencing, 3D ball-endmill scallop surfacing, and thread milling, four open research frontiers remain: (1) Off-axis undercuts and 3+2 / simultaneous 5-axis indexing for features inaccessible along $\pm Z$ without T-slot/dovetail cutters or rotary trunnions; (2) Coupled frequency-domain regenerative chatter stability lobes and thin-wall workpiece compliance [1] beyond static cantilever cutter deflection; (3) Joint 3D stepover-and-z freeform trajectory optimization over high-resolution multi-dexel fields; and (4) Instrumented physical Haas VMC machining trials measuring spindle load, cycle time, and surface finish R_a.

Conclusion. We challenged the manual paradigm of 3-axis CNC programming with DiffCAM, unifying bi-level multi-setup TAMP over a 673-record clean-room tool catalog and physics-based feeds_and_speeds with classical CAM trajectory initialization, smooth-to-hard differentiable cutting simulation, and trajectory-policy co-optimization. Across 7 experimental suites—culminating in 10/10 gouge-free multi-tool machining of arbitrary TITAN-1M–10M STEP solids (41.6% mean cycle-time reduction)—DiffCAM demonstrates that subtractive 3-axis CNC milling can approach the automated, upload-and-cut accessibility of 3D printing.

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TABLE IV

Exp4 Independent High-Resolution Verification, 6-Controller Audit, and Haas Mini Mill NGC Shop Packet Summary Across All 8 Held-Out Test Parts. Audited across 6 dialects (48/48 pass) via a 0.125 mm-step posted_evaluator (0.25 mm voxels) and containerized CAMotics v1.2.0.

Haas Prog.	Case ID	CAM Family	Stock (mm)	T_{nom} (s)	T_{DiffCAM} (s)	$-\Delta T$ (%)	Gouge (mm^3)	Res. (mm^3)	6-Dialect	CAMotics STL	STL Vol. (mm^3)
O01001	face-test-00	Facing Boss	34×24×8	79.92	32.85	58.9%	0.00	0.00	6/6 Pass	Watertight (74.9k△)	4245.9
O01002	face-test-01	Facing Boss	32×24×8	77.68	30.34	60.9%	0.00	0.00	6/6 Pass	Watertight (72.6k△)	4370.6
O01003	pocket-test-00	Pocket Island	32×28×8	117.24	96.26	17.9%	0.00	0.00	6/6 Pass	Watertight (107.9k△)	5034.5
O01004	pocket-test-01	Pocket Island	34×26×8	89.94	62.01	31.0%	0.00	0.00	6/6 Pass	Watertight (102.1k△)	5514.2
O01005	adapt-test-00	Adaptive Trochoid	38×30×8	188.64	123.36	34.6%	0.00	0.00	6/6 Pass	Watertight (129.3k△)	6675.4
O01006	adapt-test-01	Adaptive Trochoid	36×32×8	280.60	198.18	29.4%	0.00	0.00	6/6 Pass	Watertight (127.1k△)	7060.9
O01007	contour-test-00	Contour Boss	32×26×8	107.81	59.82	44.5%	0.00	0.00	6/6 Pass	Watertight (76.1k△)	4143.0
O01008	contour-test-01	Contour Boss	32×24×8	88.60	54.19	38.8%	0.00	0.00	6/6 Pass	Watertight (73.2k△)	4204.8
Mean / Pass	8 Held-Out Test Parts (36.2% Agg. Time Saved)			**128.80**	**82.13**	**39.5%**	**0.00**	**0.00**	**48/48**	**8/8** Watertight	$\Delta V = 0.000\%$

TABLE V

Exp6 End-to-End Automated Multi-Tool TAMP Pipeline from Raw .stl Input Across Four Parts and Three Alloys. Total times include 6.0 s physical ATC swaps (Tn M6, G43 Hn); all Haas NGC programs (O02001–O02004) pass chronological vise-jaw, holder, rapid, and gouge audits.

Haas Prog.	Part ID (.stl)	Alloy	Auto-Selected Tools / Ops	ATC (s)	T_{naive} (s)	T_{same} (s)	T_{opt} (s)	$-\Delta T_{\text{total}}$ / $-\Delta T_{\text{cut}}$	Opt h_{ex}	Safety / Gouge
O02001	tamp-flanged-bracket	Al 6061-T6	4 T/6 Ops (⌀10, ⌀6, Spot, ⌀4 G83)	18.0	109.11	106.84	106.48	2.4% / 2.9%	75.6%	PASS / 0.00 mm^3
O02002	tamp-sensor-housing	Al 6061-T6	4 T/4 Ops (⌀10, ⌀6, Spot, ⌀5 G83)	18.0	76.95	76.73	72.54	5.7% / 7.5%	88.8%	PASS / 0.00 mm^3
O02003	tamp-manifold-plate	Al 7075-T6	4 T/5 Ops (⌀10, ⌀6, Spot, ⌀3.5 G83)	18.0	100.00	96.76	93.88	6.1% / 7.5%	76.4%	PASS / 0.00 mm^3
O02004	tamp-stepped-mount	Steel 1018	4 T/5 Ops (⌀10, ⌀4, Spot, ⌀6 G83)	18.0	112.32	99.00	**71.07**	**36.7%** / **43.7%**	66.3%	PASS / 0.00 mm^3
Mean / Pass	4 Multi-Tool .stl Parts (673-Record Catalog)		**4.0** T/**5.0** Ops	**18.0**	**99.59**	**94.83**	**85.99**	**12.8%** / **15.4%**	**76.8%**	**4/4** PASS

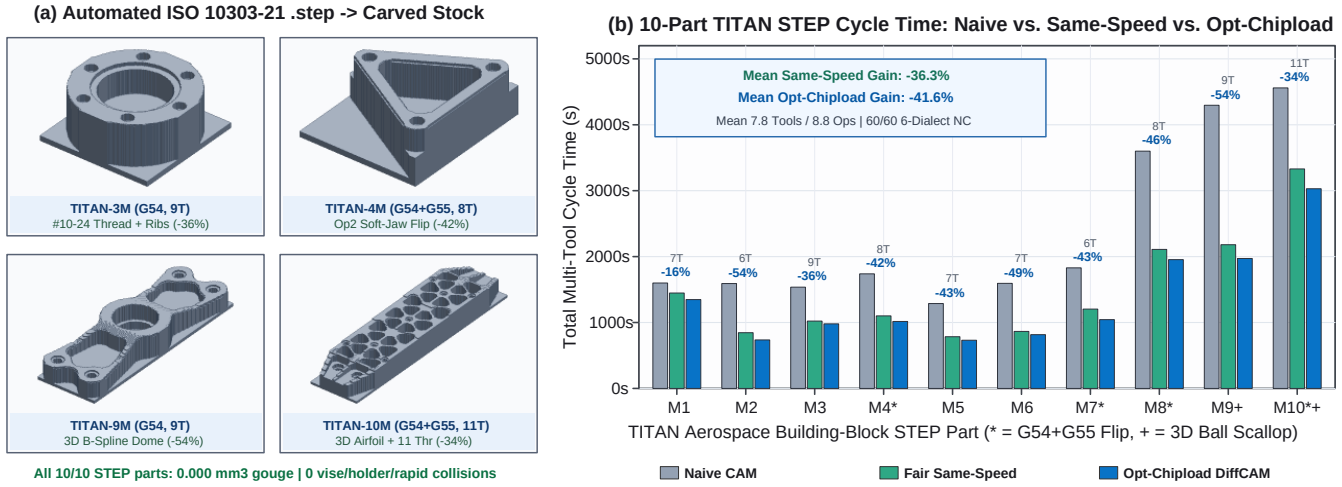


Fig. 8. Exp7 end-to-end arbitrary ISO 10303-21 .step benchmark across all 10 TITAN-1M–10M parts. (a) 4×-MSAA isometric renderings of carved 6061-T6 stock solids produced automatically from raw .step B-Rep inputs on TITAN-3M (9 tools, #10-24 climb thread milling + radial ribs), TITAN-4M (8 tools, dual-setup G54+G55 soft-jaw flip), TITAN-9M (9 tools, 3D B-spline dome finished via inverse-offset 1/4″ ball-endmill scallop passes), and TITAN-10M (11 tools, 13 operations, dual-setup 3D B-spline airfoil + 11 threads). (b) Total multi-tool cycle time across all 10 parts comparing Naive CAM, Fair Same-Speed, and Optimal-Chipload DiffCAM (**−41.6**% mean cycle-time reduction, **10/10** safety pass, **0.000** mm^3 gouge).

TABLE VI

Exp7 Arbitrary ISO 10303-21 .step Benchmark Across All 10 TITAN-1M–10M Aerospace Building-Block Parts (Al 6061-T6, US Imperial Shop Tool Crib). Setup column indicates auto-selected $SO(3)$ orientation and single (G54) vs. dual (G54+G55 M00 flip) workholding; all 10 parts achieve 0.000 mm^3 gouge, zero collisions, and 6/6 dialect compilation (103,155 total Haas NGC lines).

Part (.step)	Setup / WCS	Part Dims (mm)	Recognized B-Rep Features & Topologies	Tools / Ops	Plan (s)	T_{naive} (s)	T_{same} (s)	T_{opt} (s)	$-\Delta T_{\text{same}}$	$-\Delta T_{\text{opt}}$	Gouge (mm^3)	Haas Lines
Titan-M1	$+Z$ (G54)	101.6×48.3×19.1	Stepped pocket, 7 holes/c-bores/c-sinks, #47 pilot	7 T/7 Ops	12.1	1599.6	1447.6	1347.8	9.5%	15.7%	0.000	7,891
Titan-M2	$+Y \rightarrow +Z$ (G54)	127.0×49.5×19.1	Open boss, angled slots, 6 holes/c-bores, 16 chamfers	6 T/7 Ops	9.5	1591.4	845.2	735.9	46.9%	**53.8%**	0.000	4,743
Titan-M3	$+Y \rightarrow +Z$ (G54)	49.5×49.5×19.1	Radial ribs, 7 holes/c-bores, 1 c-sink, 6×#10-24 thr	9 T/9 Ops	12.7	1536.5	1021.8	979.6	33.5%	36.2%	0.000	9,590
Titan-M4	$+Z$ (G54+G55)	56.8×49.5×19.1	Dual-setup island pocket, 3 c-bores/thr, 9 chamfers	8 T/10 Ops	17.3	1738.2	1100.4	1016.0	36.7%	41.5%	0.000	10,586
Titan-M5	$+Y \rightarrow +Z$ (G54)	49.5×49.5×19.1	Hex flange, 5 holes/c-bores/threads, 10 chamfers	7 T/8 Ops	10.0	1287.7	784.4	731.2	39.1%	43.2%	0.000	6,921
Titan-M6	$+Y \rightarrow +Z$ (G54)	49.5×49.5×19.1	Stepped circular boss, 9 holes, 4 c-sinks, 20 chamfers	7 T/7 Ops	11.7	1594.1	865.0	815.8	45.7%	48.8%	0.000	6,554
Titan-M7	$+Z$ (G54+G55)	111.1×49.5×19.1	Dual-setup webbed housing, 4 holes/c-bores, 30 chamf	6 T/8 Ops	14.9	1829.2	1203.7	1043.6	34.2%	43.0%	0.000	6,609
Titan-M8	$+Z$ (G54+G55)	138.4×49.5×19.1	Dual-setup multi-pocket bracket, 2 c-bores/c-sinks, 8 ch	8 T/10 Ops	22.7	3599.8	2109.1	1954.0	41.4%	45.7%	0.000	14,186
Titan-M9	$+Y \rightarrow +Z$ (G54)	152.4×48.2×19.1	6× 3D B-spline dome (1/4″ Ball), ⌀31.8 bore, 5 c-bores	9 T/9 Ops	28.5	4296.1	2180.3	**1971.4**	**49.2%**	**54.1%**	0.000	16,148
Titan-M10	$+Y \rightarrow +Z$ (G54+G55)	203.2×48.3×21.5	Dual-setup 8× 3D airfoil (1/4″ Ball), 11 holes/threads	11 T/13 Ops	30.6	4559.2	3330.2	3029.5	27.0%	33.5%	0.000	19,927
Mean / Pass	10/10 Arbitrary STEP Solids (4× G54+G55 Flips, 2× 3D Ball Scallop, 60/60 6-Dialect NC)			**7.8** T/**8.8** Ops	**17.0**	**2363.2**	**1488.8**	**1362.5**	**36.3%**	**41.6%**	**0.000**	**10,316**

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